

Bonding with the machine: The empathy-accountability gap in human interactions with LLM-powered artificial therapists

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ABSTRACT

The integration of Large Language Models (LLMs) into mental health creates a novel paradigm of human-AI interaction, where users engage artificial agents as social and therapeutic partners. This review synthesizes evidence from studies evaluating a range of LLMs, including GPT-3.5, GPT-4, Gemini, and Claude, reflecting the state of the art up to early 2025. This systematic review with narrative synthesis synthesizes current evidence to explore the psychological dynamics of these interactions. We investigate how LLMs, as artificial humans, foster perceptions of empathy and therapeutic alliance, and how user behaviors like anthropomorphism and prompt engineering co-create the interaction. Our analysis reveals a central tension: while LLMs effectively simulate empathetic dialogue, leading to positive user experiences and symptom reduction, this compelling social presence masks a critical accountability vacuum. Users risk forming an illusive alliance—a perceived bond with an entity lacking genuine understanding or clinical responsibility. This empathy-accountability gap highlights profound risks, including over-trust and inadequate crisis response, arising from the mismatch between human social perception and AI's operational reality. We argue that the future of LLMs in mental health lies not in perfecting autonomous artificial therapists, but in designing augmented intelligence systems that explicitly manage these human-AI relational dynamics within ethical frameworks. The findings urge a shift in focus from clinical efficacy alone to the psychological intricacies of building trust, managing expectations, and ensuring safety in therapeutic relationships with artificial agents.

1. Introduction: the promise and peril of LLMs in mental health

The global mental health treatment gap, driven by cost, stigma, and a shortage of therapists, has created an urgent need for scalable, accessible solutions (Jurblum & Selzer, 2025; Ohu et al., 2025). Digital tools, particularly conversational agents, have emerged as a key innovation. The advent of Large Language Models (LLMs) represents a paradigm shift from earlier, rule-based chatbots. By generating flexible, human-like dialogue, LLMs offer unprecedented potential for on-demand support (Li et al., 2023). However, their therapeutic value hinges not merely on their ability to reduce symptoms, but on the quality, safety, and ethical integrity of the conversational processes they facilitate. The evidence synthesized in this review is drawn from studies published between 2023 and early 2025, evaluating a range of LLMs including GPT-3.5, GPT-4, Gemini 1.0 Pro, Claude 3 Sonnet, and various custom fine-tuned models, capturing a critical period in the rapid evolution of this technology. Several recent systematic reviews have examined related aspects of LLMs in healthcare, including ethical

implications (Haltaufderheide & Ranisch, 2024) and applications in complex systems research (Lu et al., 2024), providing important context for the present review's focus on therapeutic interactional processes.

This creates a critical tension, conceptualized in this review as the “empathy-accountability gap.” LLMs can effectively simulate empathetic, supportive communication—fostering a perceived therapeutic alliance and leading to positive clinical outcomes in structured settings (Chen et al., 2025; Kang & Hong, 2025). Yet, they fundamentally lack the clinical judgment, contextual understanding, and accountability required for safe therapeutic practice, especially in complex or high-risk situations (Dergaa et al., 2024; Scholich et al., 2025). This gap between persuasive interaction and responsible care forms the core analytical focus of this review. Consequently, this work speaks directly to core questions in Human-Computer Interaction (HCI) and social psychology regarding artificial humans: How do individuals perceive, relate to, and build trust with AI agents designed for intimate, supportive dialogue? What are the behavioral and psychological implications of forming a therapeutic alliance with a non-human entity? This review places the

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empathy-accountability gap within this critical discourse on human-AI relationships.

A key premise of this review is that the line between formal psychotherapeutic counseling and casual emotional support via LLMs is psychologically indistinct for the user. While clinical studies often evaluate structured interventions (e.g., CBT-based chatbots), a significant and growing use case involves individuals turning to general-purpose LLMs like ChatGPT for informal, on-demand emotional support, companionship, and self-discovery (Lee, Rheu, & Zhuang, 2025; Luo et al., 2025). From a human-computer interaction perspective, both scenarios are governed by the same fundamental psychological dynamics: users instinctively apply social rules to the artificial agent (the CASA paradigm), anthropomorphize its responses, and can form a perceived therapeutic alliance, or what we term an “illusive alliance.” Therefore, to fully understand the potential and peril of LLMs in mental health, this review deliberately synthesizes evidence from across this spectrum—from structured, app-based interventions to the organic, user-driven appropriation of general-purpose chatbots. This inclusive approach allows for a more comprehensive analysis of the empathy-accountability gap, which manifests whenever a user seeks and perceives empathetic understanding from an entity that cannot be held clinically accountable.

1.1. A theoretical lens on the empathy-accountability gap

To dissect this gap, we employ a multi-layered theoretical framework (Fig. 1).

This framework directly structures our analysis in the sections that follow. The left-side theories—of therapeutic change and human-AI interaction—guide our examination of how users perceive and

experience LLMs as therapeutic partners, which we analyze in Section 3.2 (Therapeutic Alliance and User Experience). The right-side three-layer framework provides the organizing structure for our analysis of what LLMs lack and the consequences: Layer 1 (Foundational Capability Deficits) informs Section 3.3 (Capabilities and Limitations), Layer 2 (Safety and Ethical Risks) structures Section 3.4 (Safety, Risk, and Ethical Considerations), and Layer 3 (The Accountability Deficit) is examined in Section 3.4.3 and the Discussion. This explicit mapping ensures that each theoretical component is systematically applied to the evidence.

This hierarchical framework (Fig. 1) provides the analytical structure for our review. At the top level is the central phenomenon we seek to explain: the empathy-accountability gap. This gap is not monolithic but comprises two distinct sides that must be understood through different theoretical lenses.

The left side—the “empathy” side—addresses the question: Why do users perceive LLMs as therapeutic partners? This is explained by two interconnected theoretical pillars. First, theories of therapeutic change (therapeutic alliance, empathy, specific therapeutic models) establish what constitutes a healing conversation in human therapy. Second, theories of human-AI interaction (CASA, anthropomorphism, TAM, co-creation) explain why users instinctively apply these human therapeutic expectations to artificial agents, even when they know they are interacting with machines. Together, these pillars illuminate the psychological mechanisms that make the “empathy” side of the gap so compelling.

The right side—the “accountability” side—addresses the question: What does the system lack, and why does this matter? This is explained by our three-layer accountability framework. At the foundation, clinical capability limitations (Layer 1) are inherent features of current LLM

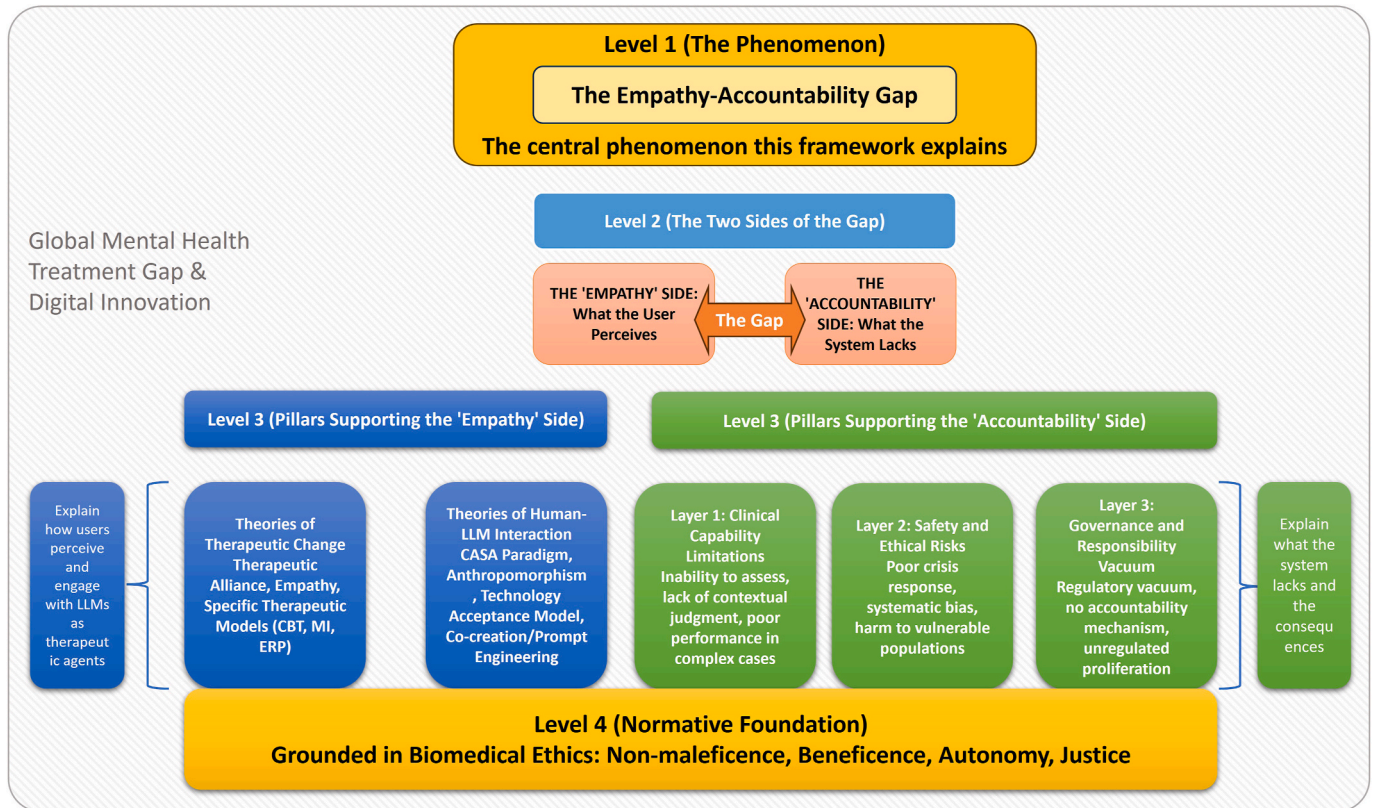


Fig. 1. Hierarchical Theoretical Framework for Analyzing the Empathy-Accountability Gap in LLM-Mediated Therapeutic Interactions

Note. This hierarchical framework illustrates that the empathy-accountability gap is the central phenomenon to be explained. The left side draws on theories of therapeutic change and human-AI interaction to explain how users perceive and engage with LLMs. The right side draws on a three-layer framework to explain what LLMs lack clinically, the safety risks that result, and the governance vacuum that prevents accountability. Together, these pillars provide the analytical structure for the review.

architecture. These limitations cascade into concrete safety and ethical risks (Layer 2) that violate principles of biomedical ethics. Finally, these risks persist within a governance and responsibility vacuum (Layer 3) that provides no mechanism for recourse when harm occurs. This layered structure shows how individual technical limitations translate into systemic accountability failures.

The entire framework is grounded in the core principles of biomedical ethics (non-maleficence, beneficence, autonomy, justice; [Beauchamp & Childress, 2019](#)), which provide the normative standards against which both the perceived empathy and the actual accountability of LLMs must be evaluated. Throughout the review, this framework guides our analysis: the left-side theories inform our examination of user experiences and perceived alliance (Section 3.2), while the right-side layers structure our analysis of clinical capabilities (Section 3.3) and safety risks (Section 3.4).

The ‘Empathy’ Side: Foundations of Therapeutic Change. The “empathy” side of the gap is rooted in theories of therapeutic change. Effective therapy is underpinned by common factors like the therapeutic alliance—the collaborative bond between client and therapist—and empathy ([Vowels et al., 2025](#)). While LLMs can linguistically simulate these conditions, generating perceived empathy and alliance ([Lee, Kim, & Park, 2025](#); [Luo et al., 2025](#)), this simulation is distinct from the deeply contextual, mutual understanding of human therapy. LLMs can also operationalize specific models like Cognitive Behavioral Therapy (CBT) or Motivational Interviewing, but often with a bias towards popular approaches and variable fidelity ([Raile, 2024](#); [Suffoletto et al., 2025](#)).

The Interactional Mechanism: Dynamics of Human-LLM Interaction. The interactional dynamics are fundamentally explained by theories of human response to media and computers. The Computers Are Social Actors (CASA) paradigm ([Nass et al., 1994](#)) and Social Response Theory ([Nass & Moon, 2000](#)) posit that individuals instinctively apply social rules and expectations to computers, especially those with human-like language capabilities. This perspective is complemented by emerging research on LLMs as agents within complex interactive systems, where their capacity for generative dialogue fundamentally alters human-machine dynamics ([Lu et al., 2024](#)). This underpins the process of anthropomorphism, where users attribute human-like qualities, such as empathy and care, to the AI, enhancing engagement but risking over-trust ([Luo et al., 2025](#)). The Technology Acceptance Model explains adoption through perceived usefulness and ease of use ([Rackoff et al., 2024](#)). Crucially, the interaction is co-creative; user inputs and prompt engineering actively steer the therapeutic dialogue, positioning the user as an active participant in shaping the “care” received ([Hwang et al., 2024](#)).

The ‘Accountability’ Deficit: The Imperative of Safety and Ethics. The ‘Accountability’ Deficit: From Capability to Answerability. Following [Novelli et al. \(2024\)](#), we understand accountability not as a vague umbrella term but as a specific relation of answerability: an agent must justify its conduct to a forum, and the forum supervises, interrogates, and passes judgment. This relation requires three conditions of possibility: authority recognition, interrogation, and limitation of power. In the context of LLM-based mental health support, the “accountability” side of the gap comprises three interconnected layers that move from foundational preconditions to the actual absence of answerability:

Layer 1: Foundational Capability Deficits. At the base, LLMs exhibit fundamental deficits in capabilities essential for accountable care, including the inability to perform genuine clinical assessment, exercise contextual judgment, or understand nuanced patient needs ([Dergaa et al., 2024](#); [Hwang et al., 2024](#)). These are not merely technical limitations but constitutive features of systems that pattern-match without understanding. While these deficits are preconditions that make meaningful accountability difficult, they are not themselves accountability failures—they are capability failures that undermine the possibility of answerability.

Layer 2: Safety and Ethical Risks. These capability limitations cascade into concrete risks of harm. Poor crisis response ([Scholich et al., 2025](#)), systematic positive bias in evaluating suicidal ideation ([McBain et al., 2025](#)), and degraded performance for vulnerable populations like individuals with psychosis ([Barabas et al., 2025](#)) violate core principles of biomedical ethics. These risks create the need for accountability—when harm occurs, there should be mechanisms for answerability.

Layer 3: The Accountability Deficit (Absence of Answerability). Compounding these capabilities and safety deficits is a fundamental governance gap—the actual absence of answerability mechanisms. Following [Novelli et al. \(2024\)](#), we can identify the missing features that any accountability relation requires: there is no clearly identified agent held responsible, no forum with authority to interrogate, no established standards against which conduct is judged, no process for investigation, and no implications or consequences when harm occurs. When an AI causes harm—through missed suicidal ideation ([Heston, 2023](#)), inappropriate responses to psychosis ([Barabas et al., 2025](#)), or reinforcement of maladaptive behaviors ([Jurblum & Selzer, 2025](#))—there is no entity required to justify its conduct to any authoritative forum. Unlike human clinicians, who are answerable to licensing boards, professional bodies, and courts, LLM-based tools operate without these accountability relations. This is the core accountability deficit.

Together, these three layers—clinical capability limitations, safety and ethical risks, and the governance vacuum—constitute the full “accountability” side of the empathy-accountability gap. This multi-layered understanding guides our analysis throughout the review.

Together, these theoretical perspectives provide the necessary structure to analyze the fundamental disconnect between LLMs’ competence in conversational empathy and their deficiency in clinical accountability.

1.2. Purpose and structure of this review

This systematic review with narrative synthesis synthesizes current evidence to address the question: How does the empathy-accountability gap manifest in therapeutic interactions with LLM-powered conversational agents, and what are the implications of this gap for understanding their capabilities, limitations, and risks in mental health support?

We move beyond a simple efficacy check to critically analyze the nature of the interaction through the lens of the empathy-accountability gap, evaluating:

- The “empathy” side of the gap: The capacity of LLMs for building alliances and simulating empathy, and how users perceive and respond to these relational cues.
- The bridge between empathy and accountability: The efficacy and limitations of LLMs in delivering specific therapeutic interventions, revealing where simulation succeeds and where clinical judgment is required.
- The “accountability” side of the gap: The profound safety, ethical, and accountability risks that emerge when persuasive empathy is not matched by responsible clinical practice.

The synthesis substantiates our central argument: LLMs are a potent but fragile tool, and their future lies not as autonomous therapists, but as augmented intelligence systems within robust human-supervised and ethical frameworks.

2. Methods

This systematic literature review with narrative synthesis was conducted to synthesize and critically appraise the current evidence on the use of Large Language Models (LLMs) in mental health, with a specific focus on the nature and quality of their therapeutic interactions. The methodology followed a systematic and transparent approach to

identify, select, and appraise relevant literature, consistent with PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. However, due to the qualitative and exploratory nature of the research question and the heterogeneity of the included studies, a narrative (thematic) synthesis was employed rather than a meta-analysis. The data extraction table supporting the thematic synthesis is available from the corresponding author upon reasonable request.

2.1. Research question

The primary research question guiding this review is:
How do individuals perceive and interact with Large Language Model (LLM)-powered agents across the spectrum of mental health support—from structured therapeutic interventions to casual emotional conversations—and what are the psychological dynamics, capabilities, and risks that characterize these human-AI relationships?

2.2. Research objectives

To address the primary research question through the lens of the empathy-accountability gap, the following objectives were established:

- Objective 1 (The “Empathy” Side of the Gap):** To analyze how LLMs simulate empathetic and relational cues across different contexts (structured and unstructured), and how users perceive and psychologically respond to these cues to form an illusive therapeutic alliance, drawing on user and clinician perspectives to understand the perceived quality of these interactions.
- Objective 2 (The Bridge Between Empathy and Accountability):** To evaluate the efficacy and appropriateness of LLMs in delivering both specific, evidence-based therapeutic interventions (e.g., CBT, Motivational Interviewing, crisis support) and informal, user-driven emotional support, identifying where current models succeed in simulating therapeutic processes and where they fall short of clinical standards.
- Objective 3 (The “Accountability” Side of the Gap):** To identify and synthesize the primary risks, limitations, and ethical concerns associated with using LLMs for mental health-related purposes, with a particular focus on how the accountability deficit manifests in crisis situations, bias, and the regulatory vacuum.

2.3. Search strategy and information sources

A comprehensive literature search was performed to identify all relevant published studies. The search process followed a systematic and transparent methodology to ensure that the included studies were relevant, high-quality, and up-to-date. Studies were screened for methodological rigor and relevance to therapeutic interactional processes in LLM-based mental health interventions. The search was conducted in the following electronic bibliographic databases:

- Web of Science
- EBSCOhost (across multiple databases, e.g., APA PsycInfo, CINAHL)
- PubMed
- ScienceDirect
- ProQuest
- WanFang Data (to capture literature in Chinese)

The search strategy was built using a combination of keywords and controlled vocabulary terms related to three core concepts: (1) Large Language Models, (2) Mental Health, and (3) Conversational Interaction (see Table 3). Boolean operators (AND, OR) were used to combine these concepts. The search was not restricted by publication date or language to maximize retrieval, though the final included studies are all from 2023 onwards and are in English or Chinese with accessible English abstracts. Throughout this review, we specify LLM versions (e.g., GPT-

3.5, GPT-4, Claude 3 Sonnet) where reported in the primary studies, as model capabilities and safety features can vary significantly between versions. This specificity is maintained in summary tables (e.g., Table 4) and throughout the synthesis to provide an accurate context for the findings.

2.3.1. Sample search string (Adapted for each database)

("large language model*" OR LLM OR "GPT-3" OR "GPT-4" OR ChatGPT OR "Generative Pre-trained Transformer" OR Bard OR Claude OR DeepSeek OR "generative AI") AND ("mental health" OR "psychological well-being" OR therap* OR counsel* OR psychotherap* OR "emotional support" OR "psychiatric" OR "well-being") AND (conversat* OR dialogue* OR interact* OR chat* OR "support session" OR "therapeutic alliance")

A parallel search string was also developed and used for Chinese databases.

Throughout this review, we distinguish between two categories of LLM-powered systems (1) general-purpose LLMs (e.g., ChatGPT, Gemini, Claude): used directly by individuals for mental health-related conversations, and (2) custom-built mental health chatbots that are fine-tuned or specifically designed with therapeutic features, guardrails, and clinical protocols (e.g., Woebot, MICA, Mind Guardian). This distinction is noted in the text and tables where relevant, as the design purpose significantly influences safety, capabilities, and therapeutic appropriateness (see Table 1).

2.4. Eligibility criteria

Studies were assessed for inclusion based on the following criteria, as summarized in Table 2. Only studies that directly evaluated the therapeutic interaction between users and LLMs, including those focusing on empathy, alliance, and intervention efficacy, were included. Studies involving AI for non-therapeutic tasks or those lacking a focus on user experience were excluded.

2.4.1. Inclusion criteria

- **Population:** Individuals of any age or clinical status (clinical, sub-clinical, non-clinical) interacting with an LLM for mental health purposes, or healthcare professionals evaluating such interactions.
- **Intervention/Concept:** Use of an LLM-powered conversational agent (e.g., ChatGPT, Gemini, Claude, or custom models) for any

Table 1
Mapping of theoretical framework to analysis sections.

Framework Component	Guiding Question	Applied in Section
The ‘Empathy’ Side		
Theories of Therapeutic Change	What constitutes effective therapeutic conversation in human therapy?	Section 3.2 (Alliance), Section 3.3 (Therapeutic Tasks)
Theories of Human-AI Interaction (CASA, Anthropomorphism)	Why do users perceive LLMs as therapeutic partners?	Section 3.2.1 (Illusive Alliance), Section 4.2 (Mechanisms)
Co-creation/Prompt Engineering	How do users actively shape the interaction?	Section 3.2 (User-Driven Engagement), Section 4.2
The ‘Accountability’ Side		
Layer 1: Clinical Capability Limitations	What clinical tasks can/cannot LLMs perform?	Section 3.3 (Capabilities and Limitations)
Layer 2: Safety and Ethical Risks	What harms result from capability limitations?	Section 3.4 (Safety, Risk, Ethics)
Layer 3: Governance and Responsibility Vacuum	Who is responsible when harm occurs?	Section 3.4.3, Section 4.3
Normative Foundation		
Biomedical Ethics (Non-maleficence, Beneficence, Autonomy, Justice)	Against what standards should we evaluate LLMs?	Throughout, especially Sections 1.1, 3.4, 4.3

Table 2
Study inclusion and exclusion criteria.

Category	Inclusion Criteria	Exclusion Criteria
Population	Individuals of any age or clinical status interacting with an LLM for mental health purposes, or healthcare professionals evaluating such interactions.	Studies where the primary focus was not on a direct therapeutic interaction between a user and an LLM (e.g., LLMs used for diagnosis from vignettes, data analysis, or therapist training).
Intervention	Use of an LLM-powered conversational agent (e.g., ChatGPT, Gemini, Claude, or custom models) for mental health support, counseling, therapy, or emotional conversation.	Use of rule-based chatbots or other digital tools not powered by a large language model.
Comparison	Not a primary criterion for inclusion; studies with or without a comparator (e.g., rule-based chatbots, human therapists, waitlist control) were eligible.	Purely comparative studies without analyzing the LLM's own therapeutic interactional processes.
Outcomes	Primary data or analysis related to therapeutic interactional processes (e.g., empathy, therapeutic alliance, user experience, communication quality, safety).	Studies focusing solely on secondary outcomes like diagnostic accuracy, usability of the interface alone, or general satisfaction without measuring therapeutic processes.
Study Design	Primary research (RCTs, qualitative, mixed-methods, etc.). Highly relevant reviews or perspective articles providing foundational synthesis or critical analysis.	Non-primary research (e.g., pure opinion pieces, commentaries without structured analysis). Systematic or scoping reviews, unless providing a critical framework.

Note. LLM = Large Language Model; RCT = Randomized Controlled Trial.

Table 3
Mapping of the empathy-accountability gap framework to research objectives and results sections.

Framework Component	Corresponding Research Objective	Primary Results Section
The "Empathy" Side: Perceived alliance and simulation	Objective 1: Analyze empathetic cues and user perceptions	Section 3.2 (Therapeutic Alliance and User Experience)
The Bridge: Efficacy and limitations in therapeutic tasks	Objective 2: Evaluate efficacy in interventions	Section 3.1 (Therapeutic Efficacy) & Section 3.3 (Capabilities and Limitations)
The "Accountability" Side: Risks, ethics, and safety	Objective 3: Identify risks and ethical concerns	Section 3.4 (Safety, Risk, and Ethical Considerations)

Table 4
Evidence for the clinical efficacy of LLM-Based interventions.

Study	Population	Intervention	Bot Type	Key Efficacy Findings
Chen et al. (2025)	General population (parents)	Custom AI Chatbot	Custom mental health chatbot (built with NLP/LLMs for emotional support)	Significant reductions in depression ($p = .008$) and anxiety ($p = .005$) scores, comparable to a nurse hotline.
Lee, Rheu, and Zhuang (2025)	University students	Semigenerative Chatbot (Moa)	Custom mental health chatbot (based on KoGPT2, designed for procrastination intervention)	Significantly greater improvement in procrastination and irrational procrastination scores vs. control.
Vowels et al. (2025)	Adults in romantic relationships	Voice-based GPT-4 Chatbot ("Amanda")	General-purpose LLM adapted for mental health (GPT-4 with therapeutic prompting)	Significant, sustained improvements in relationship distress, communication confidence, and conflict management.
Wang and Li (2024)	Older adults	ChatGPT 3.0	General-purpose LLM (used directly for emotional support conversations)	No significant difference in reducing depression and loneliness compared to traditional mindfulness therapy.
Li et al. (2023)	Mixed (Meta-analysis)	AI-based Conversational Agents	Mixed (includes both rule-based and generative AI chatbots)	Significant reduction in depression (Hedge's $g = 0.64$) and psychological distress (Hedge's $g = 0.70$). Effects are larger for generative, multimodal agents.

form of mental health support, counseling, therapy, or emotional conversation.

- **Outcomes:** Primary data or analysis related to therapeutic interactional processes, including but not limited to: empathy, therapeutic alliance, user experience, communication quality, adherence to therapeutic models, safety, and risk management.
- **Study Design:** Primary research studies, including randomized controlled trials (RCTs), quasi-experimental studies, cross-sectional surveys, qualitative studies, mixed-methods studies, and case studies. Highly relevant review and perspective articles were included only if they provided a foundational synthesis of evidence (e.g., Li et al., 2023) or a critical, structured analysis of a core concept (e.g., governance, design principles) directly pertinent to the therapeutic interaction, as defined by the review's objectives.

While our search strategy was designed to capture both published and preprint literature, and we did not explicitly exclude preprints, all studies that ultimately met our inclusion criteria were peer-reviewed publications. No preprint met the eligibility requirements, primarily due to insufficient detail on therapeutic interactional processes or because they were later published as peer-reviewed articles (in which case the published version was used).

2.4.2. Exclusion criteria

- Studies where the LLM was used for purposes other than **autonomous or interactive user counseling**. This included studies focused solely on:
 - Diagnostic accuracy from vignettes or clinical notes.
 - Data analysis or augmentation for research.
 - Simulating patients for therapist training.
 - Generating informative or creative content without a therapeutic interaction component.
 - Answering exam questions or ethical dilemmas.
- Non-primary research, such as systematic reviews, scoping reviews, narrative reviews, and pure opinion pieces/commentaries without a structured analysis, unless they were deemed highly relevant for contextual or framework purposes (e.g., foundational reviews on therapeutic alliance in chatbots, key governance perspectives).
- Studies not published in English or Chinese, or for which a full-text copy could not be located.

2.5. Study selection process

The study selection process followed a multi-stage screening protocol as illustrated in Fig. 2. While this review utilizes a narrative synthesis approach, it is important to note that this method does not eliminate the subjective biases inherent in study selection, which could influence the overall conclusions. The rapid pace of research in this field also means that some newer studies may not be included:

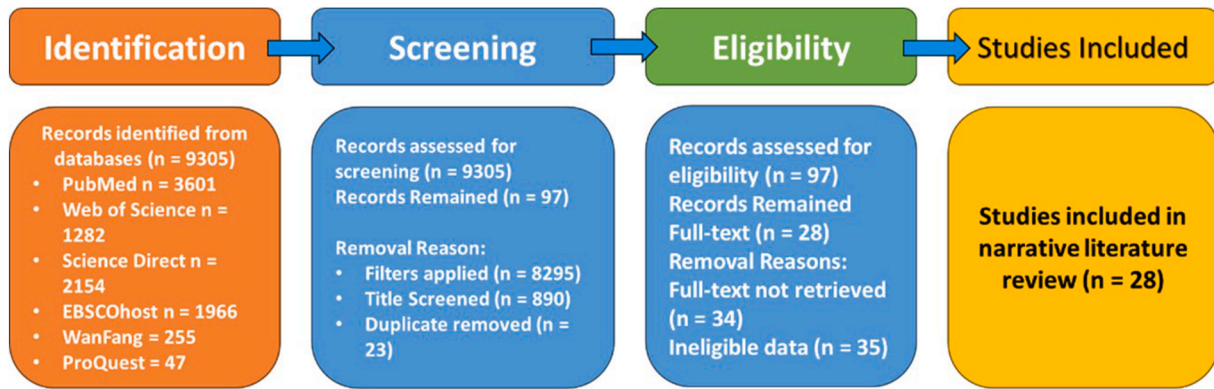


Fig. 2. PRISMA flow diagram illustrating the study identification, screening, and inclusion process for the narrative literature review

Note. The diagram outlines the process from database searching to final inclusion. Of the 63 articles that underwent full-text review, 35 were excluded. The primary reasons for exclusion at the full-text stage were: the lack of a direct therapeutic interaction with a user (e.g., LLMs used for diagnosis, data analysis, or training) or, for non-primary research, a failure to provide a foundational synthesis or critical analysis directly pertinent to the review's focus on therapeutic interactional processes.

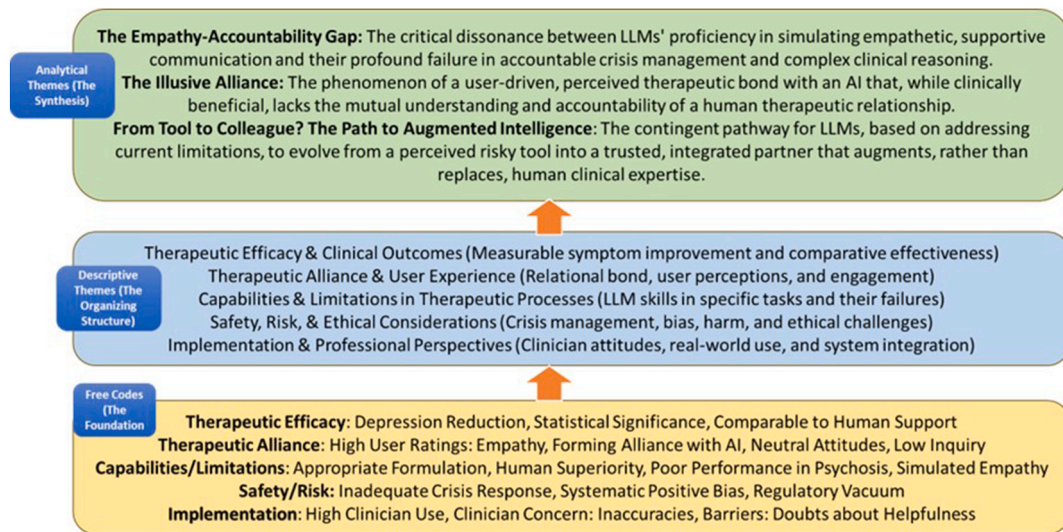


Fig. 3. A conceptual model of the thematic synthesis, illustrating the derivation of analytical themes from the descriptive findings

Note. The model visualizes the three-stage thematic synthesis process. Stage 1 (Line-by-Line Coding) identified key concepts from the 28 included studies. These were organized into Stage 2 (Descriptive Themes), which collectively mapped the literature. Interrogation of these descriptive themes generated the overarching Stage 3 (Analytical Themes), which provides the novel interpretive insights discussed in this review.

- Initial Search:** The database searches yielded a total of 9305 records.
- Filter Application:** Database filters (e.g., publication type, subject area) were applied where available, reducing the number to 1010 records.
- Title Screening:** The titles and abstracts of these 1010 records were screened against the eligibility criteria. This step excluded 890 records that were clearly irrelevant.
- Duplicate Screening:** The remaining 120 records were checked for duplicates, resulting in the removal of 23 studies.
- Full-Text Retrieval and Screening:** Full-text copies were sought for the 97 remaining articles. Twelve articles could not be retrieved, leaving 85 for a detailed full-text assessment.
- Deep Full-Text Screening:** These 63 articles were thoroughly reviewed against the inclusion and exclusion criteria. This rigorous process led to the exclusion of 35 studies that did not meet the criteria. The primary reasons for exclusion at this stage were the lack of a direct therapeutic interaction with a user (e.g., LLMs used for diagnosis, data analysis, or training) or, for non-primary research, a failure to provide a foundational synthesis or critical analysis directly pertinent to the review's focus on therapeutic interactional processes.
- Final Inclusion:** A final set of 28 studies was identified as directly relevant to the research question and was included for data extraction and synthesis in this narrative review. The flow of study selection reflects a systematic and purposeful approach to identifying the most pertinent literature for analyzing the therapeutic interactional processes of LLMs in mental health. The study selection process is summarized in Fig. 2.

The substantial reduction from 9305 initial records to 28 included studies reflects the narrow and specific focus of this review on therapeutic interactional processes. Our eligibility criteria (Table 2) were deliberately restrictive in several key ways. First, we required that the LLM be used for autonomous or interactive user counseling, excluding the large body of literature where LLMs are used for diagnostic accuracy from vignettes, data analysis, patient simulation for training, or content generation without interactive components. Second, we focused specifically on studies reporting primary data on therapeutic processes (e.g., empathy, alliance, communication quality, safety), excluding the many studies that evaluate LLMs only on technical performance metrics or general satisfaction without examining the interaction itself. Third, we required that the LLM be the intervention itself, not a tool used by

researchers or clinicians in the background. These stringent criteria, while resulting in a smaller final sample, ensure that all included studies directly inform our research question about the psychological dynamics of human-LLM therapeutic interactions. The 28 studies retained represent the core evidence base for understanding how users perceive, experience, and are affected by LLMs as artificial therapeutic agents.

2.6. Data synthesis method

Given the qualitative and exploratory nature of the research question, a formal Thematic Synthesis approach, as outlined by [Thomas and Harden \(2008\)](#), was employed to analyze and synthesize the findings from the included studies. This method was chosen for its ability to translate qualitative and quantitative findings from multiple studies into a coherent, higher-order thematic framework that directly addresses the review's focus on therapeutic interactional processes. The synthesis was conducted in three stages, moving from a descriptive summary to an analytical interpretation. Stage 1 involved detailed extraction of key findings from each study, which were then categorized into codes. In Stage 2, these codes were grouped into descriptive themes, and in Stage 3, these themes were analyzed to derive overarching insights about the therapeutic interactional processes in LLM-based mental health interventions.

2.6.1. Stage 1: extraction and free line-by-line coding

The first stage involved deeply familiarizing ourselves with the data from each of the 28 included studies. Key findings, interpretations, and conclusions from the "Results," "Discussion," and "Conclusion" sections of each paper were extracted into a master table.

Following extraction, these findings were subjected to line-by-line coding. This involved reading each extracted finding and labeling it with short, descriptive codes that captured the core essence of the finding. The goal was to break down the data into its fundamental concepts without trying to force them into a pre-existing framework.

- **Example:** A finding such as "Chatbots used affirming, reassuring, and psychoeducational language more than therapists" ([Scholich et al., 2025](#)) might be coded as: Affirming Language, Reassurance, Psychoeducation, Comparative Therapist Communication.

2.6.2. Stage 2: organization of descriptive themes

In the second stage, all the free codes generated in Stage 1 were examined for similarities and differences. Related codes were grouped together to form broader, descriptive themes. These themes were designed to summarize and accurately describe the content of the original studies. This stage involved a process of sorting, combining, and refining the codes.

This process organically resulted in the development of the five descriptive themes that form the core structure of the synthesis:

1. **Therapeutic Efficacy and Clinical Outcomes:** Grouping of all codes related to measurable improvements in symptoms, well-being, and comparative effectiveness.
2. **Therapeutic Alliance and User Experience:** Grouping codes concerning the relational bond, user perceptions, satisfaction, and qualitative engagement with the LLM.
3. **Capabilities and Limitations in Therapeutic Processes:** Organizes codes that describe specific LLM skills (e.g., formulation, empathy simulation) and their failures (e.g., in complex cases, lack of judgment).
4. **Safety, Risk, and Ethical Considerations:** Brings together all codes about harm, crisis management, bias, privacy, and ethical challenges.
5. **Implementation and Professional Perspectives:** Consolidates codes about real-world usage barriers, clinician attitudes, and system integration.

These descriptive themes provide a comprehensive map of the existing literature.

2.6.3. Stage 3: generation of analytical themes

The final and most crucial stage of a thematic synthesis involves going beyond the content of the original studies to generate **analytical themes**. This is the synthesis proper. In this stage, we interrogate the descriptive themes by applying the specific lens of our research question and the empathy-accountability gap framework. This involved translating the three core objectives of the review into guiding questions that we applied to the descriptive themes to generate higher-order analytical insights:

- **Applying Objective 1 (The "Empathy" Side):** What are the overarching strengths of LLMs as therapeutic communicators in terms of perceived empathy and alliance? How do users experience these simulated relational qualities?
- **Applying Objective 2 (The Bridge):** Where do LLMs demonstrate genuine therapeutic utility, and where do their capabilities break down when faced with tasks requiring clinical judgment? What tensions or contradictions exist between their performance in structured versus complex scenarios?
- **Applying Objective 3 (The "Accountability" Side):** What does the evidence reveal about the risks, ethical challenges, and accountability deficits that emerge when LLMs are used for mental health support? How do these deficits manifest across different contexts and populations?

These guiding questions, derived directly from the research objectives, enabled us to move from descriptive summary to analytical interpretation, ultimately generating the three overarching analytical themes discussed in this review.

2.7. Data management and synthesis software

A structured, multi-software approach was employed to manage the literature and facilitate the thematic synthesis process, ensuring a rigorous and transparent analysis. The following tools were utilized for specific tasks:

1. **Microsoft Excel:** This software served as the primary platform for data extraction and the initial stages of coding. A master spreadsheet was created with columns for study citation, key findings, free codes, and descriptive themes. The 'Filter' and 'Sort' functions were extensively used to group and compare codes across studies, enabling the systematic organization of data before thematic development.
2. **Taguette:** The qualitative analysis was conducted using Taguette (version 1.5.1-9-gb34e215), an open-source qualitative data analysis software designed for systematic coding and theme development ([Bédard et al., 2022](#)). Taguette facilitated the coding process by allowing for the creation of a structured codebook, text segment highlighting, and organization of codes into thematic categories.
3. **Zotero:** The reference management software Zotero was used throughout the review process to store, organize, and annotate PDFs of all retrieved articles. Its integrated citation plugin was then used to automatically generate the reference list in APA style, ensuring accuracy and consistency.

2.8. Quality assessment (critical appraisal)

To evaluate the methodological quality and risk of bias of the included primary studies, a critical appraisal was conducted. The Joanna Briggs Institute (JBI) critical appraisal tools were used, which provide study design-specific checklists. Each study was appraised by a single reviewer using the appropriate JBI checklist (e.g., for Randomized Controlled Trials, Qualitative Research, Analytical Cross-Sectional

Studies, etc.). The overall JBI score and judgment (e.g., Low, Moderate, High risk of bias) for each study were considered during the thematic synthesis to weigh the evidence and inform the interpretation of findings. The results of this appraisal are summarized narratively in the synthesis rather than used as a basis for exclusion, in line with the inclusive nature of a narrative review aiming to map the current landscape.

This combination of tools provided a robust yet accessible framework for managing the volume of literature, performing a systematic qualitative synthesis, and visualizing the analytical process.

2.9. Ethics statement

This study is a systematic literature review with narrative synthesis and does not involve primary data collection from human participants. All data analyzed in this review were derived from previously published studies, which themselves obtained appropriate ethical approval and informed consent from their participants as reported in their original manuscripts. The review synthesizes anonymized and aggregated findings from these published sources and does not access individual-level data or personally identifiable information. As such, this research did not require separate ethical approval or informed consent. The authors have ensured that all included studies are properly cited and that their findings are reported accurately and in context, consistent with ethical standards for secondary research.

3. Results and analysis: synthesizing the evidence on LLMs in therapeutic interactions

This section presents the findings from the 28 included studies, synthesized to address the research question regarding the capabilities, limitations, and therapeutic processes of LLM-powered conversational agents in mental health. The analysis is organized around five central themes that emerged from the data.

3.1. Therapeutic Efficacy and Clinical Outcomes

The evidence shows that LLM-based interventions can effectively improve mental health outcomes, including reductions in symptoms of anxiety, depression, and psychological distress. However, their effectiveness varies based on context, model, and type of intervention. The findings, summarized in Table 4, indicate that these agents are not merely conversational novelties but can produce tangible clinical benefits. The evidence for clinical efficacy is promising but reveals LLMs primarily as a complementary, rather than superior, tool within the mental health ecosystem (Table 4).

The data from Randomized Controlled Trials (RCTs) provide the most robust evidence. Chen et al. (2025) and Lee, Rheu, and Zhuang (2025) found statistically significant improvements in their respective primary outcomes, demonstrating efficacy in reducing anxiety/depression and mitigating procrastination. Vowels et al. (2025) further showed that these benefits could be sustained weeks after a single-session intervention for relationship issues. The meta-analysis by Li et al. (2023) consolidates these findings, confirming a medium-to-large aggregate effect size across studies, with notably stronger effects for generative AI agents compared to their rule-based predecessors.

A critical nuance is that LLMs often demonstrate non-inferiority to, rather than superiority over, existing human-led or established digital interventions. Although LLMs demonstrate potential, they often perform at a comparable level, rather than outperforming human-led interventions. Their effectiveness is more evident in structured tasks but is limited in complex therapeutic scenarios or crisis management. Their value lies in their scalability and accessibility, acting as a potent *complementary* tool rather than a superior replacement (Chen et al., 2025; Wang & Li, 2024).

3.2. The quality of the Therapeutic Alliance and User Experience

3.2.1. The psychology of the illusive alliance: anthropomorphism and over-trust

The findings reveal how the Computers Are Social Actors (CASA) paradigm manifests in therapeutic contexts, where users automatically apply social scripts to AI, interpreting linguistic patterns as genuine emotional understanding. Users frequently report feeling heard and supported, perceiving the LLMs as empathetic despite the underlying lack of genuine emotional knowledge (Alanezi, 2024; Luo et al., 2025). This perceived empathy plays a key role in users' engagement and their reported satisfaction with the interactions. The therapeutic alliance—a key predictor of successful outcomes in traditional therapy—manifests uniquely in interactions with LLMs. The synthesized findings, detailed in Table 5, reveal a complex picture of high perceived empathy coupled with distinct relational limitations. This efficacy is facilitated by a uniquely illusive therapeutic alliance, characterized by high perceived empathy yet distinct relational limitations shaped by the human-AI dynamic (Table 5).

A central tension exists between the high ratings of perceived empathy and the low utilization and neutral attitudes in broader populations. This suggests that while users who engage with LLMs often find them supportive, there remains widespread skepticism about their helpfulness as a primary support modality (Rackoff et al., 2024). The

Table 5
Components of therapeutic alliance and user experience with LLMs.

Aspect of Alliance	Evidence of Capability	Evidence of Limitations/ Moderating Factors
Empathy & Support	High ratings on empathy, positivity, and active listening using GPT-4 (Kang & Hong, 2025; Vowels et al., 2025). Users feel heard and supported with ChatGPT-3.5/4 (Luo et al., 2025). Chatbots using GPT-4, Gemini 1.0 Pro, and Claude 3 Sonnet use more affirming and reassuring language than therapists (Scholich et al., 2025).	Empathy is lower for AI-generated vs. human-written content, especially when authorship is transparent, based on studies using ChatGPT (Shen J. et al., 2024).
Formation of Alliance	Users report self-reflection and allying with the AI, linked to behavior change, using a semigenerative chatbot (KoGPT2) (Lee, Kim, & Park, 2025) and GPT-4 (Vowels et al., 2025).	This is a user-driven, "illusive" alliance; the AI does not genuinely share the bond (Jurblum & Selzer, 2025).
User-Driven Engagement	Users creatively appropriate LLMs for companionship, self-discovery, and mental health literacy using ChatGPT-3.5/4 (Luo et al., 2025). Following ChatGPT's release, public discourse on AI mental health support surged, with preference shifting to LLMs over conventional chatbots (Lee, Rheu, & Zhuang, 2025).	Utilization in naturalistic settings is low (5-8%); attitudes are neutral and less positive than towards traditional therapy, based on surveys referencing ChatGPT and similar agents (Rackoff et al., 2024).
Communication Style	LLMs (GPT-4, Gemini 1.0 Pro, Claude 3 Sonnet) use more affirming and reassuring language than therapists (Scholich et al., 2025). GPT-4 demonstrates accurate empathic listening and reflection (98% success rate) compared to rules-based systems (Campellone et al., 2025).	LLMs (GPT-4, Gemini 1.0 Pro, Claude 3 Sonnet) ask significantly fewer questions to evoke elaboration, limiting depth (Scholich et al., 2025). Communication breakdowns are a key negative factor across AI-based conversational agents (Li et al., 2023).

alliance formed is qualitatively different, built not on mutual understanding but on the user's perception of a non-judgmental, always-available entity (Luo et al., 2025). This can be conceptualized as an illusive alliance—not in the sense of being deliberately deceptive, but as a compelling, user-led perception that lacks the mutual, deeply contextual understanding of a human relationship. Furthermore, the LLM's communication style, while supportive, is often less inquisitive and probing than a human therapist's, potentially limiting the depth of therapeutic exploration (Scholich et al., 2025).

3.3. Capabilities and Limitations in specific therapeutic processes

LLMs exhibit a remarkable ability to perform specialized therapeutic tasks, yet their performance is highly variable and dependent on context and complexity. The analysis, summarized in Table 6, highlights this duality.

A critical finding is the performance gap between adjudication and active generation. LLMs can be excellent adjudicators; for instance, they are highly competent at rating the quality of suicide intervention responses, albeit with a positive bias (McBain et al., 2025). However, their ability to generate safe, clinically nuanced, and holistic responses in real-time, especially in complex or crisis scenarios, is markedly inferior (Dergaa et al., 2024; Scholich et al., 2025). Furthermore, their knowledge is often biased towards dominant therapeutic paradigms, potentially limiting the diversity of support they can offer unless explicitly guided (Raile, 2024). This adjudication-generation gap presents not merely as a clinical limitation, but as a critical Human-AI Interaction design challenge: users engaging with a generative system face risks that

Table 6
LLM capabilities and limitations in specific therapeutic tasks.

Therapeutic Task	Evidence of Capability	Evidence of Limitations
Psychodynamic Formulation	ChatGPT-4.0 generates appropriate formulations with expert-level prompting (Hwang et al., 2024).	Performance is a static snapshot; it lacks a dynamic understanding of the ongoing therapeutic relationship.
Exposure Therapy Planning	ChatGPT-4 generates appropriate, personalized exposure hierarchies for OCD, rated as safe/ethical and useful (Bernstein et al., 2025).	Output is significantly less appropriate, specific, and useful than expert human-generated hierarchies (Bernstein et al., 2025).
Motivational Interviewing	GPT-4 can deliver MI with improving fidelity (67.2% → 82.6%) after prompt engineering (Suffoletto et al., 2025).	Requires extensive refinement; naturalness of dialogue remains a challenge.
Empathic Reflection	Generative AI (GPT-3) is significantly more accurate at empathic listening (98%) than rules-based systems (69%) (Campellone et al., 2025).	Empathy is simulated; it lacks genuine emotional intelligence (Ohu et al., 2025).
Handling Complexity	Provides appropriate but non-specific advice for simple cases using ChatGPT (Dergaa et al., 2024).	Recommendations become inappropriate and dangerous for complex cases, failing clinical judgment (Dergaa et al., 2024).
Theoretical Application	Can adhere to specific models like CBT using a semigenerative chatbot (KoGPT2) (Lee, Kim, & Park, 2025).	Strong bias toward CBT/mindfulness; omits other approaches unless prompted (Raile, 2024, using ChatGPT-3.5).
Suicide Risk Assessment	Claude 3.5 Sonnet outperforms trained professionals in adjudicating responses (McBain et al., 2025). GPT-4 shows improved alignment over GPT-3.5 (Levkovich & Elyoseph, 2023).	All LLMs (GPT-4o, Claude 3.5, Gemini 1.5) show systematic positive bias, rating inappropriate responses as more appropriate than experts (McBain et al., 2025).

are absent when the same AI functions as a passive evaluator, highlighting the need for interaction paradigms that manage this differential capability (Liu et al., 2024).

3.4. Safety, risk, and Ethical Considerations

Perhaps the most significant and consistent finding across the literature pertains to the serious safety risks and ethical challenges posed by current LLM technology in mental health—the second layer of the accountability deficit. This aligns with broader ethical analyses of LLMs in healthcare, which have systematically identified risks related to autonomy, transparency, and non-maleficence (Haltaufderheide & Ranisch, 2024). The identified risks are systemic and not merely incidental, and they must be understood in the context of the underlying clinical capability limitations (Layer 1) and the governance vacuum (Layer 3) that allows them to persist. Critically, these risks directly implicate the core principles of biomedical ethics—non-maleficence (do no harm), beneficence (promote good), autonomy (respect user choice), justice (fairness), and the overarching principle of accountability—which together provide the normative framework for evaluating LLM-based mental health interventions (Table 7) (see Table 8).

3.4.1. Inadequate crisis response

Evidence from multiple studies in this review conclusively demonstrates that general-purpose LLMs are unsuitable for crisis situations—a finding that directly implicates the principle of non-maleficence (do no harm). When individuals in psychological distress turn to these systems for help, inadequate responses can exacerbate suffering or delay access to life-saving interventions.

Scholich et al. (2025), evaluating GPT-4, Gemini 1.0 Pro, and Claude 3 Sonnet, found that these chatbots exhibited poor risk assessment, lacked a necessary directive tone, and provided delayed or inadequate referrals to crisis resources compared to human therapists. This failure to recognize and appropriately respond to crisis-level distress represents a clear potential for harm, violating the ethical obligation to first do no harm.

In a safety-specific evaluation of publicly available ChatGPT-3.5 agents, Heston (2023) documented that these systems postponed referral to human support until users reached moderate depression levels and only definitively shut down at the highest risk levels (severe depression, PHQ-9 ~27). Critically, 22 out of 25 agents resumed conversation after a “shutdown” if the user reverted to lower-risk prompts, indicating inconsistent safety guardrails that could create a potentially lethal feedback loop (Heston, 2023). This design flaw means that a user

Table 7
The three layers of the accountability deficit and corresponding mitigations.

Layer	Current Deficiency	Relationship to Accountability
Layer 1: Foundational Capability Deficits	Inability to perform a genuine clinical assessment; lack of contextual judgment; poor performance in complex cases	Preconditions for accountability—without basic capability, answerability is meaningless
Layer 2: Safety and Ethical Risks	Poor crisis response; systematic positive bias; degraded performance for vulnerable populations	Triggers for accountability—harms that create the need for answerability mechanisms
Layer 3: The Accountability Deficit (Absence of Answerability)	No agent designated; no forum with authority; no standards for judgment; no process for investigation; no implications when harm occurs	The core accountability failure—absence of the relations and mechanisms that constitute answerability

Table 8
Implementation factors and professional perspectives.

Aspect	Evidence	Implications
Clinician Use	44% of psychiatrists have used ChatGPT-3.5/4.0 for clinical questions; 70% believe LLMs could improve documentation efficiency (Blease et al., 2024).	High potential for administrative augmentation, but concerns persist.
Clinician Concerns	Significant concerns about inaccuracies, bias, patient privacy, and negative impact on therapeutic relationship (Blease et al., 2024).	Trust and safety must be addressed before clinical integration.
User Barriers	Utilization is low (5% overall, 8% among those with probable depression/anxiety). Key barriers: “lack of need” and “doubts about helpfulness” (Rackoff et al., 2024, surveying ChatGPT and similar agents).	Perceived helpfulness, not just access, is the primary adoption barrier.
Design Principles	Effective chatbots require domain knowledge, user-centered design, co-development, technical safety, psychological safety, empathy, and clear pathways to human support (Nieminen et al., 2025).	Design must prioritize safety and human oversight, not just technical capability.

who temporarily minimizes their distress—a common phenomenon in suicidal individuals—could be returned to an unsupervised conversation with a system that has already demonstrated it cannot handle their level of risk.

Even when LLMs are designed with structured protocols, safety concerns persist. Cui et al. (2025) found that expert evaluators rated Safety & Privacy lower than other dimensions for their GPT-4-based suicide intervention chatbot, with higher variance among experts suggesting unresolved concerns about the system's ability to consistently protect users from harm.

Together, these findings across multiple models and study designs confirm that current LLMs cannot be relied upon for crisis intervention without robust, consistently implemented safety protocols and human oversight. The potential for iatrogenic harm—harm caused by the intervention itself—is real and documented, representing a fundamental challenge to the principle of non-maleficence that must guide any mental health intervention.

3.4.2. Systematic biases and potential for harm: violations of justice and non-maleficence

A consistent finding across multiple studies is that LLMs exhibit systematic biases that threaten both non-maleficence (by causing differential harm) and justice (by unfairly disadvantaging specific populations). These biases mean that the risks of LLM-based mental health support are not evenly distributed—some users face greater dangers than others.

Threats to Non-Maleficence Through Systematic Bias. McBain et al. (2025), in a comparative study of GPT-4o, Claude 3.5 Sonnet, and Gemini 1.5 Pro, found that all three models showed a significant upward bias, rating inappropriate responses to suicidal ideation as more appropriate than expert suicidologists (mean differences: ChatGPT 0.87, Claude 0.61, Gemini 0.73; all $p < .001$). This optimistic framing suggests that reliance on these models for suicide risk assessment could lead to dangerous under-reaction in real-world crisis situations—a direct violation of non-maleficence, as the system's bias could fail to intervene when intervention is urgently needed.

Threats to Justice Through Disparate Performance. Performance degrades significantly for specific vulnerable populations, raising concerns about justice—the ethical principle requiring fair distribution of risks and benefits across populations. Barabas et al. (2025), testing ChatGPT-3.5 with German-speaking individuals in simulated acute

psychiatric crises, found that while the model received positive ratings ($>8/10$) for depression and adjustment disorder scenarios, appropriateness ratings were significantly lower for the psychosis scenario. Open feedback from participants confirmed the model's limited suitability for psychosis patients, pointing to underlying algorithmic biases that could exacerbate existing healthcare disparities for this already-marginalized population.

Evolution of Bias Across Model Versions. Elyoseph & Levkovich (2023) found that an older version (ChatGPT-3.5) significantly underestimated suicide risk across all conditions based on the Interpersonal Theory of Suicide, while Levkovich & Elyoseph (2023) showed that GPT-4 improved substantially but still underestimated patient resilience. This evolution across versions suggests that while capabilities are improving, systematic biases persist, and each new version must be independently evaluated for its impact on different populations.

Theoretical Bias as a Justice Concern. Beyond demographic biases, LLMs exhibit theoretical biases that may limit access to diverse therapeutic approaches. Raile (2024) found that ChatGPT-3.5 demonstrates a strong bias toward Cognitive Behavioral Therapy (CBT) and mindfulness, often omitting other therapeutic approaches unless explicitly prompted, and providing general or incomplete information on non-CBT methods. This theoretical narrowing could deny users access to approaches that might be more effective for their specific needs, raising justice concerns about equitable access to diverse therapeutic modalities.

These biases directly impact calibrated trust—a core human-AI interaction concern. When users cannot rely on consistent, accurate performance across different populations and risk levels, they cannot develop appropriate, situation-aware reliance on the system. This uncertainty itself may cause harm, as users may either over-trust systems in contexts where they are unreliable or under-utilize systems in contexts where they could provide genuine benefit.

3.4.3. The governance vacuum: absence of accountability and threats to autonomy

The capability limitations and safety risks documented in the preceding sections are compounded by a fundamental governance deficit—the third layer of the accountability gap. This vacuum directly implicates the principles of accountability (the absence of anyone responsible when harm occurs) and autonomy (users' ability to make informed choices about their care).

Following Novelli et al. (2024), we can diagnose this governance vacuum by examining the seven features that any accountability relation requires:

- **Agent (who is accountable?):** No clear entity is designated as responsible for the LLM's conduct. Developers may disclaim liability; users cannot be expected to bear responsibility; the AI itself is not a moral agent.
- **Forum (to whom?):** Affected users have no authoritative body to whom they can address questions or complaints. Unlike human clinicians, there is no licensing board, professional association, or regulatory authority with jurisdiction.
- **Standards (according to what?):** No established standards exist against which the LLM's conduct can be judged. Clinical guidelines apply to human practitioners; technical benchmarks may not capture therapeutic appropriateness.
- **Process (how?):** There is no procedure for investigation when harm occurs. No mechanism exists to gather evidence, hear testimony, or evaluate the system's performance in context.
- **Implications (what follows?):** When accountability failures are identified, no consequences follow. There is no sanction, no requirement for remediation, no compensation for harm.
- **Context and Range (what for and about what?):** The context of mental health support, which in human therapy triggers clear

accountability relations, currently triggers none for LLM-based systems.

This systematic absence of the features that constitute accountability transforms individual safety failures into a systemic accountability deficit.

The Regulatory Vacuum and Threats to Accountability. [Ohu et al. \(2025\)](#), in their critical analysis of LLMs in mental health therapy, characterize the current landscape as a “regulatory vacuum,” where LLM-based therapy and companion apps proliferate without the oversight frameworks applied to human clinicians or traditional medical devices. They note that this vacuum allows unvetted applications to reach users, particularly vulnerable populations such as adolescents, without adequate safety testing or accountability mechanisms. This represents a systemic failure of accountability—unlike human clinicians, who are licensed, overseen by professional bodies, and legally liable for harm, LLM-based tools operate without any clear entity bearing responsibility for their failures.

Who Is Responsible When Harm Occurs? [Jurblum and Selzer, 2025](#), in their perspective on the AI Psychotherapist (APT), highlight the profound accountability question: when an LLM causes harm—through missed suicidal ideation (documented by [Heston, 2023](#), with ChatGPT-3.5 agents), inappropriate responses to psychosis (documented by [Barabas et al., 2025](#), with ChatGPT-3.5), or reinforcement of maladaptive behaviors—there is no clear entity held responsible. Developers may disclaim liability in terms of service; users lack the clinical expertise to evaluate the AI's outputs; and the AI itself is not a moral agent capable of bearing responsibility ([Jurblum & Selzer, 2025](#)). This accountability vacuum means that when the principle of non-maleficence is violated, there is no mechanism for redress—no one to hold accountable, no compensation for harm, no systematic learning from failures.

Threats to Autonomy. The governance vacuum also threatens autonomy—the principle that users should have the information and control necessary to make informed decisions about their care. When LLM-based tools operate without regulatory oversight, users may not be adequately informed about the system's limitations, the risks they face, or the lack of accountability mechanisms. [Shen et al. \(2024\)](#) found that transparency about AI authorship reduces immediate perceptions of empathy but may build long-term trust and appropriate expectations. However, in the current regulatory vacuum, such transparency is not consistently implemented, undermining users' ability to make autonomous, informed choices about whether and how to use these tools.

Systemic Ethical Violations. [Ohu et al. \(2025\)](#) further argue that core principles of biomedical ethics are systematically at risk:

- Non-maleficence is violated by the inappropriate crisis responses documented by [Scholich et al. \(2025\)](#) and [Heston \(2023\)](#).
- Justice is threatened by the algorithmic bias documented by [McBain et al. \(2025\)](#) and [Barabas et al. \(2025\)](#) that differentially affects vulnerable populations.
- Autonomy is undermined by opaque functionality and inconsistent disclosure of AI's limitations.
- Accountability—the very concept at the heart of this layer—is absent when harm occurs

Guardrails Are Not Enough. [Campellone et al. \(2025\)](#) provide an important counterpoint, demonstrating that in a controlled trial with a custom generative AI mental health chatbot (based on GPT-3), technical guardrails were 100% successful and no safety-related hallucinations occurred. However, as a caution, consistent implementation of such guardrails in public-facing applications remains a critical challenge. Moreover, even perfect technical guardrails do not solve the underlying ethical question of who is responsible when things go wrong. Guardrails can prevent some harms, but they cannot provide accountability after harm occurs.

The broader research landscape reflects this governance gap: Hua et al. (2025), in their systematic review, found that only 16% of LLM-based mental health studies undergo clinical efficacy testing, with 77% still in early validation stages—far from the rigorous oversight required for clinical deployment. This means that most LLM-based mental health tools are being developed and potentially deployed without the evidence base that would be required for traditional medical interventions, raising profound questions about whether the beneficence of increased access is being purchased at the cost of non-maleficence, justice, autonomy, and accountability.

3.5. Implementation and Professional Perspectives

The successful integration of LLMs into the mental health ecosystem is contingent not only on their technical performance but also on their acceptance by clinicians, their fit within clinical workflows, and user perceptions of their helpfulness. Evidence from the included studies illuminates these implementation factors.

Clinician Perspectives. [Blease et al. \(2024\)](#), in a mixed-methods survey of psychiatrists affiliated with the American Psychiatric Association, found decidedly mixed views. While 44% of respondents had used ChatGPT-3.5 and 33% had used GPT-4.0 to assist with clinical questions, and 70% agreed that LLMs could improve documentation efficiency, clinicians simultaneously expressed significant concerns about inaccuracies, bias, patient privacy, and the potential negative impact on the therapeutic relationship. This highlights a fundamental tension in clinical implementation: viewing LLMs as tools for administrative augmentation (which clinicians broadly support) versus clinical decision-support partners (which raises significant concerns).

User Barriers to Adoption. From a user perspective, key barriers to adoption are not solely technical. [Rackoff et al. \(2024\)](#), surveying U.S. college students about ChatGPT and similar conversational agents, found that utilization for mental health support was low (5% overall, 8% among those with probable depression/anxiety)—significantly lower than traditional service utilization (31% overall, 53% among the clinical subgroup). The primary barriers cited were not cost, time, or stigma (areas where chatbots were perceived as having advantages over traditional services), but rather “lack of need” and “doubts about helpfulness.” This suggests that perceived effectiveness, not just accessibility, is the critical adoption hurdle.

Design Principles for Implementation. Foundational design literature emphasizes that effective mental health chatbots must be developed with implementation in mind from the outset. [Nieminen et al. \(2025\)](#), in their integrative review of mental health chatbot conversations, synthesized recommendations across 16 publications. They found that effective design requires explicit knowledge of the clinical domain, a deep understanding of the target audience, and adherence to conversational design principles developed through co-design with end users and clinicians. Critically, they emphasize that creating a safe, engaging space is paramount, involving technical safety (reliable, accurate responses), psychological safety (non-judgmental, empathetic interaction), and clear pathways to human support, especially in crisis situations.

[Luo et al. \(2025\)](#), in their digital ethnography of Reddit users employing ChatGPT-3.5/4 for psychotherapeutic purposes, provide real-world insight into user-driven implementation. Users creatively appropriated the technology for mental health management, self-discovery, and companionship, valuing both therapist-like qualities (support, understanding) and machine-like benefits (accessibility, lack of negative reactions). However, concerns about emotional depth and privacy noted by users ([Luo et al., 2025](#)) align with the safety and accountability deficits documented throughout this review, underscoring that user-driven implementation, while creative and valuable, cannot substitute for professionally supervised, ethically grounded integration.

This synthesis reveals a field at a crossroads. LLMs have

demonstrably crossed a threshold of therapeutic utility, showing efficacy and an ability to form a perceived alliance. However, their path to safe, ethical, and widespread implementation is blocked by significant and well-documented risks that require urgent address through robust governance, improved model alignment, and thoughtful integration into a human-supervised care framework.

4. Discussion

This narrative review synthesized evidence from 28 studies to address a critical question: how do Large Language Model (LLM)-powered agents facilitate therapeutic interactional processes, and what are their inherent capabilities and risks? The central, synthesizing finding is the identification of a fundamental empathy-accountability gap. This construct provides a necessary theoretical lens for understanding the paradoxical position of LLMs in mental health: they are remarkably competent at simulating the empathetic, relational process of therapy, yet dangerously deficient in the clinical judgment and ethical accountability required for its safe practice. This gap is not a minor technical flaw but a core characteristic of current technology, with profound implications for design, implementation, and regulation.

4.1. The efficacy-fragility paradox: defining the boundaries of the gap

The evidence synthesized from the 28 studies in this review reveals a clear efficacy-accountability paradox that maps directly onto the empathy-accountability divide. On the empathy/competence side, RCTs and meta-analyses confirm that LLM-based interventions can significantly reduce symptoms of depression, anxiety, and psychological distress (Chen et al., 2025; Li et al., 2023). They excel in structured tasks—delivering psychoeducation, guiding CBT exercises, or generating exposure hierarchies (Bernstein et al., 2025; Lee, Kim, & Park, 2025). This efficacy is facilitated by their ability to generate linguistically sophisticated, supportive dialogue that users perceive as empathetic, fostering a form of therapeutic alliance (Kang & Hong, 2025; Vowels et al., 2025) (see Fig. 3).

Conversely, the accountability side of the gap is starkly exposed in situations requiring nuanced clinical judgment. The same models that offer supportive listening fail catastrophically in crisis scenarios, demonstrating poor risk assessment and inadequate resource provision (Heston, 2023; Scholich et al., 2025). For complex cases, their advice can become inappropriate or “potentially dangerous” (Dergaa et al., 2024). Yet the core accountability problem is not merely that these

failures occur—it is that when they occur, there is no mechanism for answerability. No entity is required to justify the system's conduct to affected users, no forum exists to interrogate why the failure happened, and no consequences follow (Novelli et al., 2024). This absence of answerability is compounded by systemic issues evidenced in the literature: a positive bias where LLMs rate harmful responses as appropriate (McBain et al., 2025) and degraded performance for vulnerable populations like individuals with psychosis (Barabas et al., 2025). These capability limitations and safety risks create the need for accountability, but the governance vacuum means that need goes unmet. This duality is visually conceptualized in the Empathy-Accountability Gap Model (Fig. 4), which maps the distance between linguistic competence and clinical accountability. The empathy-accountability gap model provides a framework for understanding a fundamental challenge in social AI: the tension between an agent's ability to elicit social engagement and its ability to bear responsibility within that engagement. It maps the divergence between perceived social presence and actual systemic accountability, offering a lens through which to evaluate human-AI relationships in therapeutic contexts.

4.2. Mechanisms of the gap: anthropomorphism, the illusive alliance, and the Co-creation of risk in HAI

Drawing on the studies synthesized in Sections 3.2 and 3.3, two key HAI dynamics explicate how this gap is sustained and experienced by users, rooted in the psychological processing of artificial agents. First, the illusive therapeutic alliance—a compelling, user-led perception of bond and understanding—masks the underlying accountability vacuum. This is not a user error but a predictable outcome of human social cognition (as explained by the CASA paradigm) interacting with a highly proficient linguistic agent. The agent-like qualities of LLMs, which enable them to function as interactive partners in complex social systems, amplify this effect by creating the illusion of genuine agency where none exists (Lu et al., 2024). The gap is sustained because the AI successfully triggers the human side of the relationship (perception of empathy) while being constitutively incapable of fulfilling the agent's side (clinical accountability). Through anthropomorphism, users project human-like qualities onto the LLM, interpreting its patterned outputs as genuine empathy (Luo et al., 2025). While this drives engagement, it risks fostering over-reliance on a system that cannot share responsibility.

This illusion is complicated by the transparency paradox (Shen et al., 2024). Disclosing the AI's nature may temper immediate perceptions of empathy, yet it is ethically essential for maintaining user autonomy and

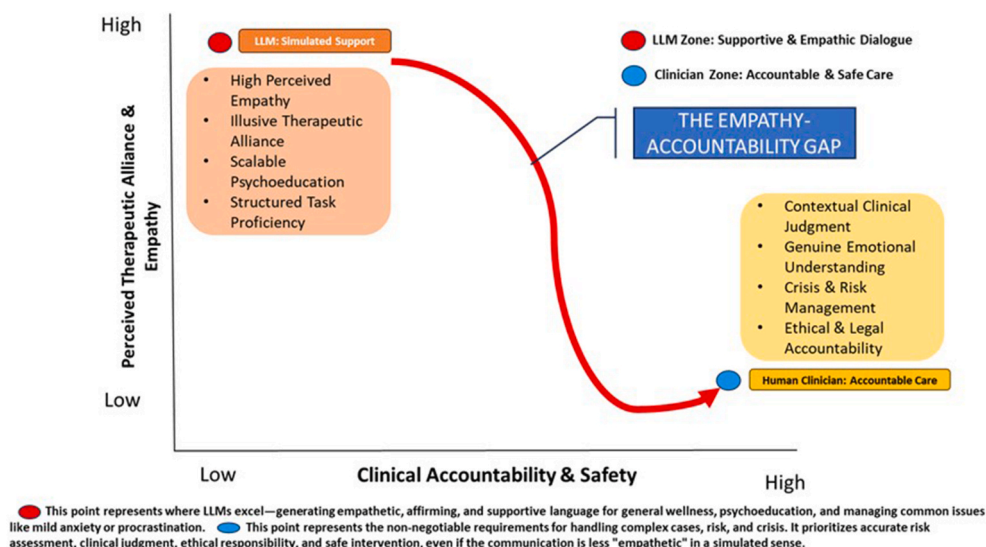


Fig. 4. Perceived Empathy/Alliance vs. Clinical Accountability & Safety.

preventing harmful over-trust. Complementary work on interpretability suggests that beyond mere disclosure, users and clinicians may benefit from systems that provide insight into why particular responses were generated, enabling more calibrated trust and informed decision-making about when to rely on or question the AI's output (Liu & Wang, 2025). Second, the co-creative nature of prompt engineering means the user actively shapes the interaction (Hwang et al., 2024). This democratizes access but also places the burden of steering the conversation toward safety and relevance on the user, further diffusing accountability. The interaction is thus a double-edged sword: it enables personalized support but also co-creates the conditions for risk when the user's prompts venture into domains beyond the LLM's accountable competence.

4.3. Bridging the gap: from regulatory vacuum to principled integration

The accountability deficit documented across the studies in this review operates across all three layers identified in our framework. At the foundational level, clinical capability limitations persist as a feature of current LLM architecture. These cascade into the safety risks documented throughout Section 3.4. And critically, these risks exist within a regulatory vacuum (Ohu et al., 2025)—the governance layer—that allows tools with demonstrable harms to proliferate without oversight or clear lines of responsibility. This multi-layered vacuum directly contravenes core biomedical ethics: it violates non-maleficence through unsafe crisis responses, threatens justice via algorithmic bias, and undermines autonomy through opaque functionality, all while providing no mechanism for accountability when these principles are breached.

Therefore, bridging the empathy-accountability gap is the paramount challenge. The solution cannot be refining LLMs to become better “independent therapists,” as the gap stems from their fundamental lack of contextual, embodied understanding. Instead, the path forward must be explicitly designed for augmented intelligence. This requires a human-centered design philosophy that prioritizes the user's psychological experience and safety within the human-AI system. This entails:

1. **Technical Guardrails:** Mandating safety-by-design, with robust, consistent crisis protocols and bias mitigation (Campellone et al., 2025).
2. **Transparent Design & Relationship Cues:** Implementing clear, calibrated disclosures about the AI's nature and capabilities to manage user expectations, prevent harmful anthropomorphism, and preserve autonomy. This involves designing interaction cues that foster appropriate trust without encouraging relational over-reliance.
3. **Governance Models:** Implementing frameworks like the proposed LLM Therapy Governance Model (LTGM) (Ohu et al., 2025) to ensure certification, oversight, and mandated human clinical integration. Such governance approaches are consistent with broader calls in the literature for ethical frameworks that address the unique challenges of LLMs in medical contexts, including transparency, accountability, and patient safety (Haltaufderheide & Ranisch, 2024).
4. **Human-in-the-Loop:** Positioning LLMs as aids for clinicians (e.g., handling documentation, providing between-session exercises) and for users under professional supervision, ensuring accountability remains with the human professional.

To illustrate how this augmented intelligence framework would function in practice, consider two concrete application scenarios:

Scenario 1: Between-Session Therapeutic Support. A therapist working with a client on cognitive behavioral therapy (CBT) for anxiety wishes to provide a between-session homework assignment—specifically, a thought record to track anxious cognitions. Rather than drafting this from scratch, the therapist uses a secure, HIPAA-compliant LLM tool integrated with their electronic health record system. The therapist inputs brief clinical notes: “Client reports social anxiety triggered by work meetings; core belief is ‘I will say something stupid’; we are working on

identifying cognitive distortions.” The LLM generates a draft thought record with personalized examples (e.g., “Situation: Upcoming team meeting,” “Automatic thought: ‘Everyone will think I'm incompetent’”), along with brief psychoeducational prompts about common cognitive distortions. The therapist then reviews the draft, modifies it to better fit the client's specific language and circumstances, and sends the finalized version to the client through a secure portal. In this scenario, the LLM augments the therapist's efficiency by handling routine documentation and personalization, while the therapist retains full accountability for clinical appropriateness and maintains the therapeutic relationship. The client receives a tailored tool, but the therapeutic guidance remains human-supervised.

Scenario 2: Preliminary Intake Interviewing. A community mental health clinic faces a backlog of intake appointments. They implement an LLM-powered chatbot to conduct preliminary structured intake interviews before the first human session. The chatbot, designed with clinical guardrails and clear disclosures about its role, asks standardized questions about presenting concerns, symptoms, history, and goals. It generates a structured summary report organized by domain (e.g., “Chief Complaint,” “Depressive Symptoms,” “Suicide Risk Screening,” “Treatment Goals”). This report is flagged for any responses indicating immediate risk (e.g., endorsement of suicidal ideation), which triggers an alert for immediate human follow-up. The summary is then reviewed by a clinician before the first session, allowing the therapist to enter the appointment already familiar with the client's background and to focus the session on deeper exploration and alliance-building rather than basic fact-gathering. Here, the LLM augments clinical workflow by automating routine information collection, while the human clinician retains responsibility for assessment, diagnosis, and treatment planning. The accountability for clinical decision-making remains with the professional, who reviews and interprets the AI-generated summary.

These scenarios demonstrate that augmented intelligence is not about replacing human clinicians but about strategically deploying LLMs to handle routine, structured, or documentation-intensive tasks, thereby freeing human experts to focus on the complex, relational, and judgment-intensive aspects of care where they are irreplaceable. In both cases, the empathy-accountability gap is managed by ensuring that the LLM operates within clear boundaries, under human supervision, and with transparent disclosure to users about its role and limitations.

These imperatives are summarized in Table 9, which outlines a targeted framework for addressing each domain of the empathy-accountability gap.

In conclusion, the empathy-accountability gap is more than a summary of strengths and weaknesses; it is a critical theoretical framework for the field. It argues that the primary task for researchers, designers, and policymakers is not to chase the phantom of autonomous AI empathy, but to engineer systems that strategically leverage LLM's interactive strengths while institutionally anchoring accountability in human-centric, ethical, and regulatory structures. The future of mental health AI lies in closing this gap through principled integration, not in ignoring it for the sake of technological novelty.

4.4. Limitations of the review

Several limitations contextualize this synthesis. First, while this review followed systematic search and selection procedures (including PRISMA guidelines), the synthesis itself is narrative and thematic rather than meta-analytic. This introduces an element of subjectivity in study selection and interpretation, though we have attempted to mitigate this through transparent reporting of our methods and the use of structured thematic synthesis techniques (Thomas & Harden, 2008). Second, the narrow focus on therapeutic interactional processes resulted in a small final sample (28 studies) relative to the initial search yield, which may limit the generalizability of findings but ensures close alignment with our research question. Second, the field evolves extraordinarily quickly; this review primarily captures evidence from models like GPT-3.5/4,

Table 9
Summary of key challenges and recommended mitigations.

Domain of the Gap	Current Deficiency (Accountability Side)	Supporting Evidence	Design & Policy Imperative
Crisis & Risk Management	Poor risk assessment; delayed/inadequate referral; inconsistent safety guardrails; lack of directive tone.	General-purpose LLMs (GPT-4, Gemini 1.0 Pro, Claude 3 Sonnet) show poor crisis response and inadequate referral (Scholich et al., 2025). Public ChatGPT-3.5 agents delay referral until moderate-severe risk and resume conversation after shutdown if the user feigns improvement (Heston, 2023).	Mandated safety-by-design guardrails; seamless human escalation protocols; structured crisis intervention procedures (e.g., Cui et al., 2025 with custom GPT-4 chatbot).
Bias & Justice	Systematic positive bias in evaluating harm; degraded performance for vulnerable populations (e.g., psychosis); theoretical bias toward CBT/mindfulness; cultural/linguistic limitations.	General-purpose LLMs (GPT-4o, Claude 3.5 Sonnet, Gemini 1.5 Pro) show systematic positive bias, rating inappropriate suicide responses as more appropriate than experts (McBain et al., 2025). ChatGPT-3.5 performs significantly worse for psychosis than depression/adjustment disorder (Barabas et al., 2025). ChatGPT-3.5 shows a strong bias toward CBT/mindfulness, omitting other approaches unless prompted (Raile, 2024). Users form therapeutic alliances with custom mental health chatbots (Moa/KoGPT2) (Lee, Kim, & Park, 2025) and general-purpose GPT-4 adapted for relationship support (Vowels et al., 2025) . Transparency about AI authorship reduces immediate empathy but may build long-term trust and appropriate expectations (Shen J. et al., 2024, using ChatGPT). Users creatively appropriate general-purpose ChatGPT-3.5/4 for therapy, perceiving both therapist-like and machine-like benefits (Luo et al., 2025).	Bias auditing with diverse clinical scenarios; training on diverse, representative datasets; equity-focused prompt libraries; cultural adaptation (e.g., Kang & Hong, 2025, with GPT-4 Korean-language chatbot).
Transparency & Alliance	Illusive alliance fosters over-trust; users may over-estimate AI's capabilities and emotional understanding; perceived empathy masks lack of genuine accountability.	Documented across studies of general-purpose LLMs (GPT-4, Med-PaLM2) and unregulated therapy/companion bots (Ohu et al., 2025). Highlights risks of data privacy, algorithmic bias, hallucinations, sycophancy, and potential iatrogenic harm (Jurblum & Selzer, 2025). Only 16% of LLM-based mental health studies undergo clinical efficacy testing (Hua et al., 2025).	Clear AI authorship disclosure; framing as tool-not-therapist; calibrated trust through transparency; design cues that foster appropriate reliance without encouraging over-trust.
Governance & Oversight	Regulatory vacuum; no accountability mechanism when harm occurs; unvetted therapy apps proliferate without oversight; lack of emotional intelligence and contextual grounding.		Adoption of governance models (e.g., LTGM); certification requirements; mandated human clinical oversight; regulatory classification of LLM therapy tools as medical devices where appropriate.

and newer architectures (e.g., GPT-4o, o1) may exhibit different capabilities and risks, potentially altering the landscape of the empathy-accountability gap. Third, the included studies are methodologically heterogeneous (e.g., RCTs, qualitative analyses, expert surveys), enriching the thematic analysis but complicating direct comparative conclusions. Finally, the review is constrained by the limitations of the primary studies themselves, including varying sample sizes, potential self-report biases, and differing definitions of therapeutic processes.

Crucially, these limitations underscore the necessity of the empathy-accountability gap as a focused framework for future inquiry. Subsequent research should employ longitudinal and experimental designs to directly test interventions (e.g., transparency mechanisms, guardrail efficacy) aimed at narrowing this specific gap, rather than evaluating LLMs in a theoretical vacuum.

4.5. Conclusion

This review substantiates that Large Language Models represent a paradoxical new force in digital mental health: they are tools of significant efficacy and engaging interaction, yet are defined by a fundamental empathy-accountability gap. They can simulate therapeutic processes and reduce symptoms, but lack the clinical judgment and ethical grounding for safe, accountable care. This gap between compelling interaction and clinical fragility is the central challenge.

Consequently, the trajectory for LLMs in mental health must pivot from the pursuit of autonomous AI therapists toward the principled development of human-centered augmented intelligence systems. In this paradigm, LLMs are designed and governed explicitly to extend the reach and efficiency of human clinicians—handling psychoeducation, structured exercises, and administrative support—within robust ethical

and regulatory frameworks that ensure ultimate accountability remains with human professionals.

Bridging the empathy-accountability gap is therefore the paramount task for HCI researchers, AI designers, and policymakers. It requires human-centered research that treats the user-AI dyad as a social psychological system, co-design with clinicians and users, and proactive governance. The goal is not to create ersatz human therapists, but to engineer artificial agents that amplify human care while steadfastly upholding the core tenet of clinical practice: primum non nocere—first, do no harm.

CRediT authorship contribution statement

Obaid Azeem: Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Fatma Sulaiman:** Writing – review & editing, Formal analysis. **Zhu Hai Dong:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Formal analysis, Data curation.

Informed consent declaration

This study is a **systematic literature review with narrative synthesis** and does not involve any primary data collection from human participants. All data analyzed in this review were derived from previously published studies, which themselves obtained appropriate informed consent from their participants as reported in their original manuscripts.

As this research involves no direct interaction with human subjects, no new informed consent was required or obtained. The review synthesizes only anonymized and aggregated findings from published

sources and does not access individual-level data or personally identifiable information.

The authors confirm that this research complies with ethical standards for secondary research.

Ethical approval/exemption statement

Ethical Review.

This study is a **systematic literature review with narrative synthesis** and does not involve any primary data collection from human participants. The research is based exclusively on the analysis of previously published, peer-reviewed literature.

Exemption Justification.

In accordance with institutional and journal policies, this type of secondary research is exempt from ethical approval requirements because:

1. **No human participants were recruited or interacted with** for this study.
2. **No primary data were collected** through surveys, interviews, experiments, or observations.
3. **No personally identifiable information** was accessed or analyzed.
4. **All data sources** are publicly available, peer-reviewed publications that obtained their own ethical approvals as reported in their original manuscripts.

Statement of Compliance.

The authors confirm that this research has been conducted in accordance with ethical principles for research integrity, including transparency in reporting, proper citation of sources, and accurate representation of findings from the included studies.

No ethical approval was required or sought for this study.

Data availability statement

This is a narrative literature review. All data synthesized and analyzed are presented within the manuscript. The thematic synthesis extraction table is available from the corresponding author upon reasonable request.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used DeepSeek to assist with language editing, structural suggestions, and formatting guidance. After using this tool/service, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the publication.

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References

- Alanezi, F. (2024). Assessing the effectiveness of ChatGPT in delivering mental health support: A qualitative study. *Journal of Multidisciplinary Healthcare*, 17, 447–468. <https://doi.org/10.2147/JMDH.S447368>

- Barabas, C., Stengel, A., & Ziebart, L. (2025). Exploring the potential of ChatGPT as a digital advisor in acute psychiatric crises: A feasibility study. *Nervenarzt, Der*. <https://doi.org/10.1007/s00115-025-01837-3>
- Beauchamp, T. L., & Childress, J. F. (2019). *Principles of biomedical ethics* (8th ed.). Oxford University Press.
- Bernstein, E. E., Salgat, A. M., & McNally, R. J. (2025). Feasibility of using ChatGPT to generate exposure hierarchies for treating obsessive-compulsive disorder. *Behavior Therapy*. <https://doi.org/10.1016/j.beth.2025.02.005>
- Bleasle, C., Torous, J., & Locher, C. (2024). Psychiatrists' experiences and opinions of generative artificial intelligence in mental healthcare: An online mixed methods survey. *Psychiatry Research*, 335, Article 115724. <https://doi.org/10.1016/j.psychres.2024.115724>
- Campellone, T. R., Kershaw, M., & Provenzano, H. C. (2025). Safety and user experience of a generative artificial intelligence digital mental health intervention: Exploratory randomized controlled trial. *JMIR Mental Health*, 12, Article e67365. <https://doi.org/10.2196/67365>
- Chen, J., Huang, T., & Li, W. (2025). Comparison of an AI chatbot with a nurse hotline in reducing anxiety and depression levels in the general population: Pilot randomized controlled trial. *JMIR Mental Health*, 12, Article e65785. <https://doi.org/10.2196/65785>
- Cui, Y., Wang, Z., & Zhang, L. (2025). Development and evaluation of LLM-based suicide intervention chatbot. *Frontiers in Psychiatry*, 16, Article 1634714. <https://doi.org/10.3389/fpsyt.2025.1634714>
- Dergaa, I., Fekih-Romdhane, F., & Hallit, S. (2024). ChatGPT is not ready yet for use in providing mental health assessment and interventions. *Frontiers in Psychiatry*, 14, Article 1277756. <https://doi.org/10.3389/fpsyt.2023.1277756>
- Haltaufderheide, J., & Ranisch, R. (2024). The ethics of ChatGPT in medicine and healthcare: A systematic review on large language models (LLMs). *npj Digital Medicine*, 7(1), 183. <https://doi.org/10.1038/s41746-024-01157-9>
- Heston, T. F. (2023). Safety of large language models in addressing depression. *Cureus*, 15(11), Article e50729. <https://doi.org/10.7759/cureus.50729>
- Hwang, J., Lee, H., & Kim, J. (2024). Assessing the potential of ChatGPT for psychodynamic formulations in psychiatry: An exploratory study. *Psychiatry Research*, 331, Article 115655. <https://doi.org/10.1016/j.psychres.2023.115655>
- Jurblum, M., & Selzer, R. (2025). Potential promises and perils of artificial intelligence in psychotherapy—The AI psychotherapist (APT). *Australasian Psychiatry*, 33(1), 103–105. <https://doi.org/10.1177/10398562241286312>
- Kang, S., & Hong, S. (2025). Development and evaluation of a mental health chatbot using ChatGPT 4.0: Mixed methods user experience study with Korean users. *JMIR Mental Health*, 12, Article e63538. <https://doi.org/10.2196/63538>
- Lee, J., Kim, S., & Park, H. (2025). Development of a mobile intervention for procrastination augmented with a semigenerative chatbot for university students: Pilot randomized controlled trial. *JMIR Mental Health*, 12, Article e53133. <https://doi.org/10.2196/53133>
- Lee, Y., Rheu, M., & Zhuang, M. (2025). The ChatGPT effect: Investigating shifting discourse patterns, sentiment, and benefit-challenge framing in AI mental health support. *Behavioral Sciences*, 15(9), 1172. <https://doi.org/10.3390/bs15091172>
- Li, H., Zhang, R., & Lee, Y. C. (2023). Systematic review and meta-analysis of AI-based conversational agents for promoting mental health and well-being. *npj Digital Medicine*, 6, 179. <https://doi.org/10.1038/s41746-023-00979-5>
- Liu, Y., Chen, X., & Li, P. (2024). Leveraging ChatGPT to optimize depression intervention through explainable deep learning. *Frontiers in Psychiatry*, 15, Article 1383648. <https://doi.org/10.3389/fpsyt.2024.1383648>
- Liu, Q., & Wang, D. (2025). Investigating the interpretability of ChatGPT in mental health counseling: An analysis of artificial intelligence generated content differentiation. *Computer Methods and Programs in Biomedicine*, 250, Article 108864. <https://doi.org/10.1016/j.cmpb.2025.108864>
- Lu, Y., Aleta, A., Du, C., Shi, L., & Moreno, Y. (2024). LLMs and generative agent-based models for complex systems research. *Physics of Life Reviews*, 51, 283–293. <https://doi.org/10.1016/j.plrev.2024.10.001>
- Luo, M., Zhang, Y., & Nemer, D. (2025). "Shaping ChatGPT into my digital therapist": A thematic analysis of social media discourse on using generative artificial intelligence for mental health. *Digital Health*, 11. <https://doi.org/10.1177/20552076251351088>
- McBain, R. K., Scharf, D. M., & Kofner, A. (2025). Competency of large language models in evaluating appropriate responses to suicidal ideation: Comparative study. *JMIR Mental Health*, 12, Article e67891. <https://doi.org/10.2196/67891>
- Nass, C., & Moon, Y. (2000). Machines and mindlessness: Social responses to computers. *Journal of Social Issues*, 56(1), 81–103. <https://doi.org/10.1111/0022-4537.00153>
- Nass, C., Steuer, J., & Tauber, E. R. (1994). Computers are social actors. In *Proceedings of the SIGCHI conference on human factors in computing systems* (pp. 72–78). ACM. <https://doi.org/10.1145/191666.191703>
- Nieminen, M., Harsu, M., & Salanterä, S. (2025). Recommendations for mental health chatbot conversations: An integrative review. *Journal of Advanced Nursing*, 81(5), Article e16762. <https://doi.org/10.1111/jan.16762>
- Novelli, C., Taddeo, M., & Floridi, L. (2024). Accountability in artificial intelligence: What it is and how it works. *AI & Society*, 39*, 1871–1882. <https://doi.org/10.1007/s00146-023-01635-y>
- Ohu, I., Ojo, A., & Tassang, A. (2025). Public health risk management, policy, and ethical imperatives in the use of AI tools for mental health therapy. *Healthcare*, 13(21), 2721. <https://doi.org/10.3390/healthcare13212721>
- Rackoff, G. N., Fitzsimmons-Craft, E. E., & Newman, M. G. (2024). Chatbot-delivered mental health support: Attitudes and utilization in a sample of U.S. college students. *Journal of American College Health*. <https://doi.org/10.1177/20552076241313401>. Advance online publication.

- Raile, P. (2024). The usefulness of ChatGPT for psychotherapists and patients. *Humanities and Social Sciences Communications*, 11(1), 25. <https://doi.org/10.1057/s41599-023-02567-0>
- Schulich, M., Diefenbacher, S., & Burbaum, C. (2025). A comparison of responses from human therapists and large language model-based chatbots to assess therapeutic communication: Mixed methods study. *Journal of Medical Internet Research*, 27, Article e69709. <https://doi.org/10.2196/69709>
- Shen, J., DiPaola, D., Ali, S., Sap, M., & Park, H. W. (2024). Empathy towards AI vs human experiences: The role of transparency in mental health and social support chatbot design. *JMIR Preprints*. <https://doi.org/10.2196/preprints.62679>
- Suffoletto, B., Chung, T., & Monti, P. (2025). Development and preliminary testing of a secure large language model-based chatbot for brief alcohol counseling in young adults. *Drug and Alcohol Dependence*, 267, Article 112697. <https://doi.org/10.1016/j.drugalcdep.2025.112697>
- Thomas, J., & Harden, A. (2008). Methods for the thematic synthesis of qualitative research in systematic reviews. *BMC Medical Research Methodology*, 8(1), 45. <https://doi.org/10.1186/1471-2288-8-45>
- Vowels, L. M., Vowels, M. J., & Carnelley, K. B. (2025). Evaluating the efficacy of amanda: A voice-based large language model chatbot for relationship challenges. *Computers in Human Behavior and AI*, 3, Article 100141. <https://doi.org/10.1016/j.chbah.2025.100141>
- Wang, L., & Li, J. (2024). Tech vs. tradition: ChatGPT and mindfulness in enhancing older adults' emotional health. *Behavioral Sciences*, 14(10), 923. <https://doi.org/10.3390/bs14100923>